Topics by Subject

Topic lists reflect the published round structure and the style of past BMT papers in the official archive; BMT does not publish a formal syllabus. Priority is my assessment.

Subject	Topic	Key ideas & techniques	Priority	Where it appears
Algebra	Polynomials & Vieta	Vieta's formulas, symmetric sums, Newton sums, root transformations, remainder/factor theorem	High	General, Algebra Focus, Guts
Algebra	Equations & systems	Substitution, symmetry, absolute value and radical equations, Simon's Favorite Factoring Trick	High	General, Guts
Algebra	Inequalities & optimization	AM-GM, Cauchy-Schwarz, completing the square, equality cases	High	Algebra Focus, Guts
Algebra	Sequences & series	Arithmetic/geometric, telescoping, linear recurrences, characteristic equations	High	General, Algebra Focus
Algebra	Functions & functional equations	Substituting special values, involutions, composition, floor/ceiling	Medium	Algebra Focus
Algebra	Complex numbers	Polar form, roots of unity, roots-of-unity filter, geometry in the complex plane	High	Algebra Focus, Guts
Algebra	Logarithms & exponents	Change of base, exponential equations	Medium	General
Algebra	Trigonometry (algebraic)	Identities, sum-to-product, Chebyshev polynomials, trig substitutions	Medium	Algebra Focus, Geometry Focus
Geometry	Triangles	Area formulas (Heron, rs, abc/4R), angle bisector theorem, Stewart, medians, centers	High	General, Geometry Focus, Guts
Geometry	Circles	Power of a point, cyclic quadrilaterals, tangents, Ptolemy, radical axis	High	Geometry Focus, Guts
Geometry	Similarity & ratios	Ceva, Menelaus, mass points, homothety	High	General, Geometry Focus
Geometry	Coordinates & vectors	Shoelace, distance from point to line, dot product, coordinate bashing	High	All
Geometry	Trigonometry in geometry	Law of sines/cosines, extended law of sines, trig Ceva	High	Geometry Focus
Geometry	Polygons & area	Regular polygons, dissection, Pick's theorem	Medium	General, Guts
Geometry	3D geometry	Volumes, cross-sections, spheres inscribed/circumscribed, space diagonals	Medium	Geometry Focus, Guts
Geometry	Transformations & advanced tools	Inversion, spiral similarity, complex/barycentric coordinates	Low	Late Geometry Focus
Discrete - Combinatorics	Counting basics	Permutations with repetition, stars and bars, complementary counting, casework	High	General, Discrete Focus, Guts
Discrete - Combinatorics	Inclusion-exclusion & bijections	PIE, derangements, lattice paths, Catalan numbers	High	Discrete Focus
Discrete - Combinatorics	Recursion	Setting up recurrences, Fibonacci-type counts, states	High	Discrete Focus, Guts
Discrete - Combinatorics	Probability & expected value	Linearity of expectation, conditional probability, geometric probability, Markov states	High	General, Discrete Focus, Guts
Discrete - Combinatorics	Pigeonhole, invariants, games	Extremal arguments, parity, colorings, winning positions	Medium	Discrete Focus, Power
Discrete - Combinatorics	Graph theory basics	Degrees, trees, paths, counting graphs	Medium	Discrete Focus, Power
Discrete - Combinatorics	Generating functions	Coefficient extraction, roots-of-unity filter	Low	Late Discrete Focus
Discrete - Number Theory	Divisibility & primes	Prime factorization, number/sum of divisors, gcd/lcm, Euclidean algorithm	High	General, Discrete Focus, Guts
Discrete - Number Theory	Modular arithmetic	Fermat/Euler, orders, Chinese Remainder Theorem, last digits	High	Discrete Focus, Guts
Discrete - Number Theory	Diophantine equations	Factoring, mod obstructions, bounding, Pell-type	Medium	Discrete Focus
Discrete - Number Theory	Arithmetic functions & valuations	phi, tau, sigma, Legendre's formula, lifting the exponent	Medium	Discrete Focus
Discrete - Number Theory	Bases & digits	Base conversion, digit sums, repeating decimals	Medium	General, Guts
Calculus	Limits	L'Hopital, Taylor expansion, squeeze, limits as Riemann sums	High	Calculus Focus, Guts
Calculus	Derivatives	Chain/product rules, implicit and logarithmic differentiation, related rates, optimization	High	Calculus Focus
Calculus	Integration techniques	Substitution, by parts, partial fractions, trig substitution, symmetry (King's rule)	High	Calculus Focus, Guts
Calculus	Advanced integrals	Differentiation under the integral sign, reduction formulas, improper integrals	Medium	Late Calculus Focus
Calculus	Series	Geometric and power series, Taylor series, convergence, summing by differentiating	High	Calculus Focus
Calculus	Applications	Area, volume of revolution, arc length, average value	Medium	Calculus Focus
Calculus	Differential equations & multivariable	Separable ODEs, basic double integrals, polar coordinates	Low	Occasional
Power Round (proof skills)	Proof techniques	Direct, contradiction, contrapositive, induction (weak and strong), extremal principle	High	Power
Power Round (proof skills)	Proof writing	State what you prove, define notation, justify each step, cite earlier parts	High	Power
Power Round (proof skills)	Reading new definitions	Work small examples as soon as a definition appears	High	Power
Power Round (proof skills)	Useful background	Sets and functions, basic group/ring ideas, graph theory, linear algebra, modular arithmetic, real analysis basics	Medium	Power
Power Round (proof skills)	Past topic example	BMT 2024's Power Round was on origami constructions (confirmed). Topics change every year and are not announced in advance	Info	Power
Guts-specific	Estimation	Fermi estimates, prime counting, logarithm approximations, orders of magnitude	Medium	Final Guts set
Guts-specific	Speed & accuracy	Mental arithmetic, squares to 30, powers of 2 and 3, common fraction-decimal conversions	Medium	Guts

Resources

Free unless noted. The official archive is the single best source: every past BMT paper with solutions.

Resource	Use it for	Where	Cost
BMT archives (2012-2025)	Real past papers and solutions for every round	berkeley.mt/archives	Free
BMT Power Round archive	Proof-round practice and a feel for topic style	berkeley.mt/archives/bmt/power-round	Free
AoPS Wiki - BMT problems and solutions	Community solutions and discussion	artofproblemsolving.com/wiki	Free
AoPS Volume 2	Core algebra, counting, number theory, geometry at General/Focus level	Art of Problem Solving	Paid
AoPS Intermediate Algebra / Counting & Probability / Number Theory	Deep subject preparation for Focus tests	Art of Problem Solving	Paid
Evan Chen - EGMO (Euclidean Geometry in Mathematical Olympiads)	Geometry Focus	Book; sample chapters on the author's site	Paid
Other tournaments' archives: SMT, HMMT, CMIMC, PUMaC	Same format (subject tests, guts, power); extra practice	Each tournament's website	Free
AP Calculus BC materials; MIT Integration Bee past problems	Calculus Focus speed and technique	College Board; MIT math website	Free
How to Prove It (Velleman) or Book of Proof (Hammack)	Power Round proof writing	Book of Proof is free online	Free / Paid